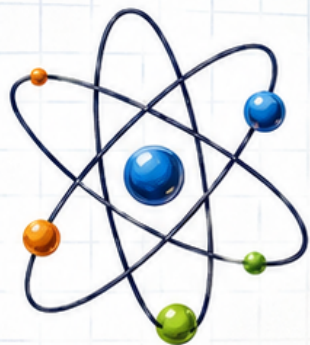
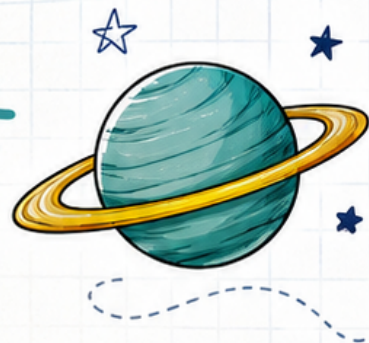




 **SAVE TIME. STAY PREPARED.**
WE'VE GOT YOUR BACK!



EMERGENCY SUB PLANS

$$E=mc^2$$

MIDDLE SCHOOL SCIENCE



**ENGAGING
MEANINGFUL
STANDARDS-
ALIGNED**



NO-PREP
Just Print
& Go!



READY-TO-USE
Complete Plans
& Materials



FOR SUB DAYS
Engaging, Clear,
& Easy to Follow



★ Created by Teachers,
for Teachers! ★



QUALITY RESOURCES. CONFIDENT CLASSROOMS. **FOCUSED ON WHAT MATTERS.**

Set this up while you are well

15 MINUTES

You are reading this because you are **well**. Spend fifteen minutes now and this pack will run itself on a morning when you are not.

The one job you have today

Print pages 3–6 and leave them in a clearly labeled folder where a substitute will find them — top drawer, on the desk, or at the front office. Everything a sub needs is in those four pages.

You do not need to print the lessons ahead of time. The sub picks one on the day and copies it.

SETTING IT UP • FIFTEEN MINUTES, ONCE

- 1 **Print the whole pack once.** Put it in a folder or binder. Write your name and room number on the cover.
- 2 **Print pages 3–6 a second time** and clip them to the front. Those are the pages a substitute reads first.
- 3 **Tell your department chair where it lives.** On a real sick day you will not be answering the phone, so somebody else needs to know.
- 4 **Cross off lessons as you use them.** There is a tracker on page 4. Two years of absences will never repeat if you keep it current.
- 5 **Optional: leave a class roster** and a note about students who need a reader, extra time, or specific seating.

Why there is no lab in this pack

Most schools will not let a substitute run a practical, and rightly so. Equipment, chemicals and safety supervision all need the regular teacher.

So none of these lessons need anything beyond a pencil. Nothing is heated, poured, cut or plugged in.

Why it is not busy work

Every lesson targets a real Grade 6–8 science idea and produces graded evidence you can put in a folder.

Away for a week? Running four of these in sequence is a defensible unit, not a holding pattern.

About the answer keys

Every key sits on the **last page of its lesson** and explains **why**, not just what. That is what lets a substitute who has not studied science since high school answer a student who asks *“but how do you know?”*

You do not need to be a science teacher

You have walked into a middle school science class. You may not have studied science since you were at school yourself. That is completely fine — this pack was written for you.

Read this and you are ready

You do not need to know any science, and you will not be running a lab. Every lesson is a paper worksheet with a complete answer key that explains the reasoning. Your job is to hand it out, keep the room settled, and check answers against the key.

DO THESE FOUR THINGS

- 1 **Pick one lesson** from the table on the next page. Any of them works for Grades 6, 7 or 8. If you cannot decide, use Lesson 06.
- 2 **Copy the student pages only.** Each lesson tells you exactly which page numbers. One per student.
- 3 **Keep the teacher page and the answer key.** Do not copy those.
- 4 **Read the teacher page once** before the bell. Two minutes, and it includes the exact words you can use to start.

Do not open the cupboards

Nothing in this pack needs equipment, and you should not hand any out. If students ask to do an experiment, the honest answer is that practicals need their own teacher present. That is a school rule almost everywhere, not you being unfair.

If they finish early

Every lesson has a challenge task on its final student page. There is also a discussion question on the teacher page you can run with the whole class for the last five minutes.

What to leave behind

Collect the sheets. Fill in the handover note on the last page of this pack — ninety seconds, and it tells the regular teacher which lesson you used, how far the class got, and anything that happened.

Leave both on the desk.

Choose your lesson

10 OPTIONS

Every lesson is standalone. There is no order and no prerequisite — pick whichever suits the class in front of you.

#	LESSON	STRAND	DIFFICULTY	MINS	STUDENT PAGES
01	Think Like a Scientist Variables & fair tests	Science practices	★	55	7–9
02	Inside a Cell Cell structure & function	Life science	★★	55	12–14
03	Push, Pull, Go Forces & motion	Physical science	★★	55	17–19
04	Reading the Periodic Table Elements & the table	Physical science	★★★	50	22–24
05	Where Does the Energy Go? Energy transfers	Physical science	★★	55	27–29
06	Break the Food Web Ecosystems & feeding relationships	Life science	★	55	32–34
07	Solid, Liquid, Gas Particle model & changes of state	Physical science	★★	55	37–39
08	Water on the Move The water cycle & weather	Earth science	★	50	42–44
09	A Restless Planet Earth's layers & plate tectonics	Earth science	★★	55	47–49
10	Do the Numbers Agree? Reading data & evaluating claims	Science practices	★★★	60	52–54

Easiest to supervise

06, 08 and 10. Diagrams and reasoning tasks where students have opinions. The room stays busy.

Best for a quiet class

02, 04 and 07. Labeling and sorting with definite answers. Fast finishers race — that is what the challenges are for.

Best for a restless class

01, 03 and 09. Argument-friendly. Fair tests, tug-of-war and earthquakes all get students talking.

Used-lesson tracker • for the regular teacher

LESSON	DATE	CLASS	LESSON	DATE	CLASS
01 — Think Like a Scientist			06 — Break the Food Web		
02 — Inside a Cell			07 — Solid, Liquid, Gas		
03 — Push, Pull, Go			08 — Water on the Move		
04 — Reading the Periodic Table			09 — A Restless Planet		
05 — Where Does the Energy Go?			10 — Do the Numbers Agree?		

The first ten minutes

SCRIPT

The first four minutes decide the next fifty. Here is a script that works even if you have never met this class.

SAY THIS AT THE DOOR

"Come in and take a seat. Bags off the desks. We are not doing a lab today – everything is on paper."

THEN, ONCE THEY ARE SEATED

"Your teacher is out today. My name is _____. This is a real science lesson, not a free period. There is a worksheet, it gets collected at the end, and your teacher will see it."

Expect the equipment question

Somebody will ask if they can do an experiment. The answer is no, and the reason is genuine: practicals need their own teacher present for safety. Say it plainly once and the class will move on.

THE SHAPE OF EVERY LESSON IN THIS PACK

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Settle, hand out sheets	Use the script above
5–12	Read the first panel together	Have a student read the explanation aloud
12–40	Students work through the tasks	Circulate. Check against the answer key
40–50	Challenge task for early finishers	Point them to the last student page
50–55	Collect sheets, pack up	Fill in the handover note

"I don't get it"

Almost always means *I don't know how to start*. Point at the worked example or diagram on the first page and ask them to describe what they can see in it.

That single move unsticks most students without you needing to explain any science.

"That's not what my teacher said"

They may well be right, and science teachers use different wording. Do not argue the point.

Say that both may be fine, ask them to write their version down, and note it on the handover sheet for the regular teacher to settle.

Copying reminder

Student pages only. Each teacher page gives the exact range. The **answer key is always the last page** of the lesson – keep it.

Think Like a Scientist

2 MINUTES

Students learn to name the three kinds of variable, spot what is wrong with an unfair test, and redesign one so it actually proves something. This is the skill every other science lesson depends on.

You do not need to know this subject

No science knowledge needed. Every task is about spotting what was changed and what should have stayed the same. The key explains each answer.

Photocopy these pages only

Pages 7–9

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Nothing else.

WHAT TO SAY TO START

"Here is an experiment somebody actually did. They changed four things at once and then announced what caused the result. Today you are going to work out why that means they proved absolutely nothing."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the variables panel together	The three names matter most
14–26	Task 1.1 — name the variables	Answers are in the key
26–40	Task 1.2 — what makes it unfair	Expect debate. Good
40–50	Task 1.3 — redesign the test	Several correct answers
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"What is the difference between independent and dependent?"

You choose the independent variable. You measure the dependent one. A memory aid: 'I change the Independent, I Depend on the result.'

"Why can you only change one thing?"

If you change two and the result changes, you cannot tell which one caused it. That is the whole idea of Task 1.2.

"Do we need to know the science to answer?"

No. Every question is about the design of the test, not about plants or batteries. Say this before they start.

If they finish early

Task 1.4 asks students to design a fair test of their own. Push early finishers to list the controlled variables they would find **hardest** to keep the same in real life.

Curriculum focus • for the regular teacher's records

Planning and carrying out investigations; identifying independent, dependent and controlled variables; evaluating experimental design. Discussion question: *"Is it ever possible to control every single variable? What do scientists do about the ones they cannot control?"*

One change at a time

16 MIN

NAME _____

CLASS _____

DATE _____

Three kinds of variable, and you must know all three

INDEPENDENT

The **one thing you change** on purpose. You pick it.

DEPENDENT

The thing you **measure** to see what happened.

CONTROLLED

Everything you deliberately **keep the same**. Usually the longest list.

Memory aid: *I change the Independent. I Depend on the result.*

Worked example

Question: does more sunlight make a plant grow taller?

Independent: hours of sunlight per day (2, 4, 6, 8)

Dependent: height of the plant in cm after 3 weeks

Controlled: same species, same pot size, same soil, same water, same temperature, same starting height

Notice the controlled list is the longest. **That is normal.** A good experiment holds many things still so that one thing can move.

Task 1.1 • Name the variables

DO

THE QUESTION

INDEPENDENT

DEPENDENT

ONE CONTROLLED VARIABLE

Does a heavier ball roll farther?

Does hotter water dissolve sugar faster?

Does fertilizer make grass grow greener?

Does a longer string make a pendulum swing slower?

What went wrong here?

20 MIN

Task 1.2 • The unfair test

DETECTIVE

Maya wanted to find out whether a new plant food makes tomato plants grow taller. Here is what she did.

Plant A — old soil, no plant food, on the windowsill, watered every day, small pot

Plant B — fresh soil, plant food added, in the greenhouse, watered twice a day, large pot

After four weeks, Plant B was 18 cm taller.

Maya concluded: "The plant food works."

a. List **four** things Maya changed between the two plants.

b. Why can she not claim the plant food caused the difference?

c. Name one other thing that could have caused Plant B to grow taller. _____

Task 1.3 • Fix it

REDESIGN

Rewrite Maya's experiment so it is a fair test.

YOUR PLAN

What she should **change**

What she should **measure**

Five things she must **keep the same**

How many plants should she use in each group, and why not just one? _____

Design your own

16 MIN

Task 1.4 • Your investigation

CREATE

Pick one of these questions, or invent your own.

Does paper airplane wing length affect how far it flies?

Does the surface a ball bounces on change its bounce height?

Does music volume affect how fast people finish a puzzle?

My question:

MY ANSWER

Independent variable • what I change

Dependent variable • what I measure

How I will measure it

Four controlled variables

Task 1.5 • The hard part

CHALLENGE

Of your four controlled variables, which would be the **hardest** to actually keep the same in real life? Explain.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. Is it ever possible to control every single variable? What should a scientist do about the ones they cannot control?

Answer key

Think Like a Scientist

ANSWERS

Task 1.1 • Naming the variables

QUESTION	INDEPENDENT	DEPENDENT	CONTROLLED (ANY ONE)
Heavier ball rolls farther?	mass of the ball	distance rolled	same ramp, same slope, same surface, same start point
Hotter water dissolves sugar faster?	water temperature	time to dissolve	same volume of water, same amount of sugar, same stirring
Fertilizer makes grass greener?	amount of fertilizer	color of the grass	same grass type, same soil, same water, same light
Longer string, slower pendulum?	length of string	time for one swing	same mass, same starting angle, same place

Watch for a common error: students often write “the plant” or “the experiment” as the dependent variable. Push them to name **a thing you could write a number or a description next to.**

Task 1.2 • Maya’s unfair test

- She changed **five** things, not four: the soil, the plant food, the location, the watering, and the pot size. Accept any four.
- Because **any one of those five could have caused the difference**, and there is no way to tell which. Her test cannot separate the plant food from everything else she changed.
- Any of: the greenhouse was warmer • the bigger pot gave more root room • fresh soil had more nutrients • twice-daily watering. All are reasonable.

Task 1.3 • The fixed experiment

Change: whether the plant gets plant food (or how much).

Measure: height in cm after four weeks.

Keep the same: soil, pot size, location, temperature, light, water amount and timing, plant variety, starting height.

How many plants? Several in each group — five or more. One plant might grow oddly for reasons that have nothing to do with the experiment. Repeating protects you from a fluke.

Grading Task 1.4

Full credit if the independent variable is something they genuinely control, the dependent variable is measurable with a number or a clear description, and at least three controlled variables are sensible. Do not mark down for an unusual question — the design is what is being assessed.

The last word

No, it is never possible. Real experiments always have variables nobody can hold still — tiny temperature shifts, differences between individual plants, the weather. Scientists deal with this by **repeating the experiment many times** and by using large groups, so random differences average out. Accept any answer reaching “repeat it” or “use a lot of them”.

Inside a Cell

2 MINUTES

Students label an animal cell and a plant cell, match each part to the job it does, and work out which parts are found in only one of the two. Highly visual and very markable.

You do not need to know this subject

Every label is on the answer key with the diagram numbers, so you can check a student's sheet at a glance without knowing any biology yourself.

Photocopy these pages only

Pages 12–14

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Colored pencils are nice for the diagrams but not required.

WHAT TO SAY TO START

"Everything alive is made of these. You are made of about thirty trillion of them. Today you are going to find out what is inside one, and what the difference is between you and a blade of grass."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the parts-and-jobs panel	Read the job list aloud together
14–28	Task 2.1 — label the animal cell	Numbers match the key
28–40	Task 2.2 — the plant cell and the difference	The key comparison
40–50	Task 2.3 — match part to job	Straight matching
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Do plants have mitochondria?"

Yes. Plants need to release energy too. Chloroplasts capture light energy; mitochondria release it. Plants have both.

"What is the difference between a cell wall and a cell membrane?"

Plants have both. The membrane is a thin flexible layer inside; the wall is a rigid outer layer made of cellulose that gives the plant its shape.

"Are these drawn to scale?"

No, and that is worth saying. Real cells vary enormously in size and most organelles are far smaller relative to the cell than in any textbook diagram.

If they finish early

Task 2.4 asks students to explain why a plant does not fall over while an animal needs a skeleton. Push early finishers to suggest what would happen to a plant cell if its cell wall were removed.

Curriculum focus • for the regular teacher's records

Structure and function of cells; comparing plant and animal cells; relating structure to function. Discussion question: "A cell is smaller than a grain of dust and does everything needed to stay alive. What does that tell you about how complicated living things are?"

The parts and their jobs

18 MIN

NAME _____

CLASS _____

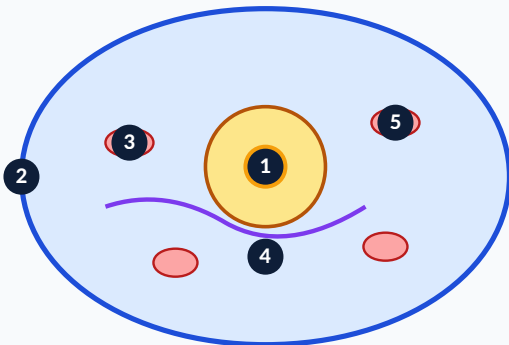
DATE _____

Every cell is a tiny factory

PART	WHAT IT DOES
Nucleus	The control center. Holds the instructions (DNA) and tells the cell what to do.
Cell membrane	A thin flexible skin. Controls what gets in and what gets out.
Cytoplasm	The jelly that fills the cell. Where most reactions happen.
Mitochondria	Release energy from food. Sometimes called the powerhouse.
Cell wall	Plants only. A rigid outer layer that gives the cell its shape.
Chloroplasts	Plants only. Capture light and use it to make food. They are what makes plants green.
Vacuole	A storage bag for water and nutrients. Huge in plant cells, tiny in animal cells.

Task 2.1 • Label the animal cell

DO



#	NAME OF THE PART	WHAT IT DOES
1		
2		
3		
4		
5		

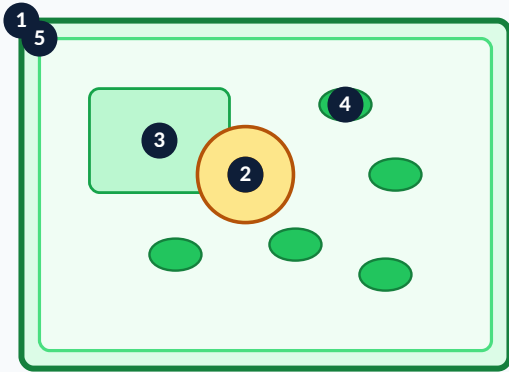
Which **two** parts from the table above would you **not** expect to find here? _____

Now the plant cell

20 MIN

Task 2.2 • Label the plant cell

DO



NAME OF THE PART

1

2

3

4

5

Which part tells you instantly that this is a plant cell and not an animal cell? _____

Task 2.3 • Same or different?

COMPARE

PART	ANIMAL CELL	PLANT CELL	BOTH
Nucleus			
Cell wall			
Cell membrane			
Chloroplasts			
Mitochondria			
Large vacuole			
Cytoplasm			

Check one column per row. Some rows have more than one correct answer — pick the **best** one.

Why the difference matters

16 MIN

Task 2.4 • Structure and job

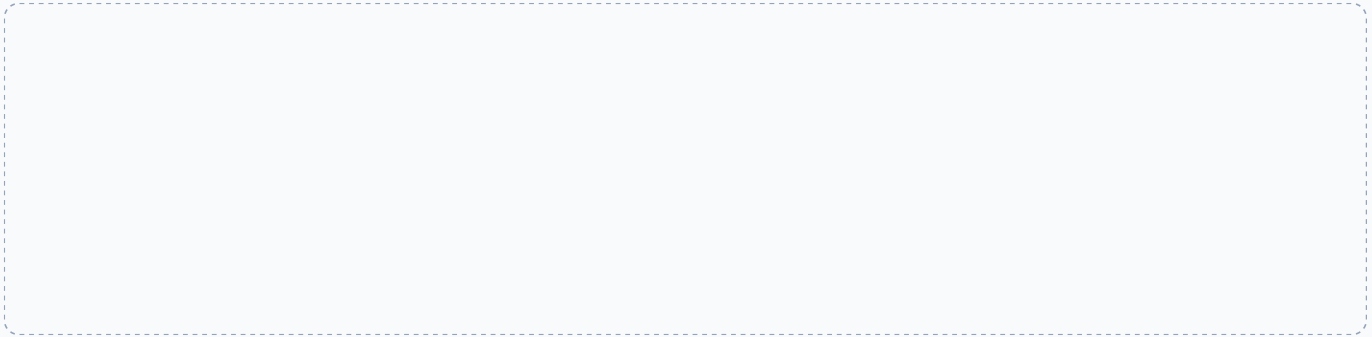
THINK

#	QUESTION	YOUR ANSWER
a	A tree stands up without any bones. Which cell part makes that possible?	
b	Why are chloroplasts green?	
c	A cell needs lots of energy. Which part would you expect to find lots of?	
d	What would happen to a plant cell if its cell wall were removed?	

Task 2.5 • Build your own

CHALLENGE

Draw a cell from an imaginary organism that lives deep underground where there is no light at all. Label three parts and explain one part you left out on purpose.



I left out _____ because _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Inside a Cell

ANSWERS

Task 2.1 • The animal cell

- 1 Nucleus — control center, holds DNA
 - 2 Cell membrane — controls what enters and leaves
 - 3 Mitochondria — release energy
 - 4 Cytoplasm — jelly where reactions happen
 - 5 Mitochondria (second one)
- Missing from an animal cell: **cell wall** and **chloroplasts**.

Task 2.2 • The plant cell

- 1 Cell wall — rigid outer layer
 - 2 Nucleus
 - 3 Vacuole — large water store
 - 4 Chloroplast
 - 5 Cell membrane — just inside the wall
- The giveaway is the **cell wall** or the **chloroplasts** — either answer is correct.

Task 2.3 • Same or different

PART	ANSWER	NOTE
Nucleus	Both	
Cell wall	Plant only	Made of cellulose
Cell membrane	Both	Plants have a membrane AND a wall
Chloroplasts	Plant only	This is why plants are green
Mitochondria	Both	Plants need to release energy too — commonly missed
Large vacuole	Plant only	Animal cells have tiny ones, so accept 'plant' or 'both' with reasoning
Cytoplasm	Both	

The mitochondria row is the one to check. Many students assume plants use chloroplasts *instead of* mitochondria. They use both — chloroplasts capture the energy, mitochondria release it.

Task 2.4 • Structure and job

- a. The **cell wall**. Millions of rigid boxes stacked together hold the tree up without any skeleton.
- b. They contain **chlorophyll**, a green pigment that absorbs light. Accept “they have a green chemical in them”.
- c. Lots of **mitochondria**. Muscle cells are a real example.
- d. It would **lose its shape** and could burst if it took in too much water. The membrane alone is too floppy to hold the shape.

Task 2.5 • what to look for

The best answer leaves out **chloroplasts**, because there is no light underground to capture. A student who reasons that far has connected structure to environment, which is the point of the lesson. Accept other answers with sound reasoning.

Push, Pull, Go

2 MINUTES

Students read force diagrams, work out the resultant force and decide what the object actually does, then apply the same idea to a tug-of-war, a parachute and a car. Very markable and surprisingly argumentative.

You do not need to know this subject

The math is subtraction. The rule is simple: balanced forces mean no change in motion, unbalanced forces mean speeding up, slowing down or turning. Every answer is in the key.

Photocopy these pages only

Pages 17–19

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil and a ruler for the force arrows.

WHAT TO SAY TO START

“Here is a question that catches almost everyone. A car is driving down the highway at a steady sixty. Are the forces on it balanced or unbalanced? Most people get this wrong, and by the end of today you will know exactly why.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the balanced/unbalanced panel	The steady-speed case is the key
14–26	Task 3.1 — read the diagrams	Subtraction plus a decision
26–38	Task 3.2 — real situations	Expect argument on the car
38–50	Task 3.3 — draw your own	Key gives what to look for
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“If the car is moving, aren't the forces unbalanced?”

No — and this is the big idea. Forces cause a **change** in motion. Steady speed in a straight line means balanced forces, even at 100 mph.

“What does N mean?”

Newtons, the unit of force. Treat it like any other unit — no need to explain where it comes from.

“Which way does the object move?”

Toward the larger force. If they are equal, its motion does not change at all — which may mean staying still, or continuing at the same speed.

If they finish early

Task 3.4 asks students to explain why a skydiver stops accelerating. Push early finishers to explain why a passenger lurches forward when a bus brakes.

Curriculum focus • for the regular teacher's records

Forces and motion; balanced and unbalanced forces; Newton's first law. Discussion question: “If you threw a ball in deep space where there is nothing to slow it down, when would it stop?”

Balanced or not?

16 MIN

NAME _____

CLASS _____

DATE _____

Forces change motion. They do not cause motion.

BALANCED FORCES

Equal and opposite. The motion does **not change**. A still object stays still. A moving object keeps going at the same speed in the same direction.

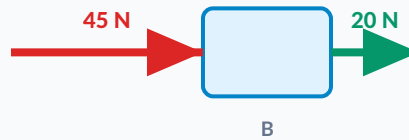
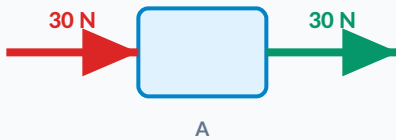
UNBALANCED FORCES

One side is bigger. The object **speeds up, slows down or changes direction**, in the direction of the larger force.

The idea that catches everybody: a car at a steady 60 mph has **balanced** forces. The engine pushes forward exactly as hard as friction and air push back. Nothing is changing, so nothing is unbalanced.

Task 3.1 • Read the diagrams

DO



BOX	RESULTANT FORCE	DIRECTION	BALANCED?	WHAT THE BOX DOES
A				
B				
C				

Task 3.1b • Name the force

DO

WHAT IS ACTING	NAME OF THE FORCE	WHAT IS ACTING	NAME OF THE FORCE
Pulls everything toward Earth		A table pushing up on a book	
Slows a box sliding across the floor		Keeps a boat floating	
Pushes back on a falling parachute		A stretched spring pulling back	

Which two of these six always act **against** motion? _____

Out in the real world

20 MIN

Task 3.2 • Balanced or unbalanced?

DECIDE

THE SITUATION

BALANCED

UNBALANCED

HOW DO YOU KNOW?

A book resting on a table

A car driving at a steady 60 mph on a straight road

A car speeding up to pass a truck

A skydiver falling at a constant speed under an open parachute

A tug-of-war rope that is not moving

A ball rolling to a stop on grass

Task 3.3 • The tug-of-war

CALCULATE

Red team pulls left with 420 N. Blue team pulls right with 380 N.

a. What is the resultant force? _____

b. Which way does the rope move? _____

c. Blue team adds one more person who pulls with 40 N. What happens now, and why? _____

d. Both teams are pulling as hard as they can and the rope is not moving. Are they using any force at all? Explain.

Task 3.3b • Quick resultants

CALCULATE

Give the size and direction of the resultant force, then say what the object does.

LEFT FORCE

RIGHT FORCE

RESULTANT

DIRECTION

WHAT IT DOES

50 N

50 N

80 N

35 N

12 N

40 N

0 N

25 N

Draw and explain

16 MIN

Task 3.4 • Draw the forces

DRAW

Draw arrows to show the forces in each case. Make the **longer** arrow the bigger force.

A parachute just after opening



A rocket lifting off



A book on a table



Task 3.5 • Explain it

CHALLENGE

a. A skydiver speeds up at first, then falls at a steady speed even though gravity is still pulling. Why does she stop speeding up?

b. When a bus brakes suddenly, standing passengers lurch forward. What is actually happening to them?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. If you threw a ball in deep space, where there is nothing at all to slow it down, when would it stop?

Answer key

Push, Pull, Go

ANSWERS

Task 3.1 • Reading the diagrams

BOX	RESULTANT	DIRECTION	BALANCED?	WHAT IT DOES
A	$30 - 30 = 0 \text{ N}$	—	Balanced	Motion does not change — stays still, or keeps moving steadily
B	$45 - 20 = 25 \text{ N}$	left	Unbalanced	Speeds up to the left
C	$40 - 15 = 25 \text{ N}$	right	Unbalanced	Speeds up to the right

For box A, “it stays still” is only **half** right — if it was already moving it keeps moving. Worth saying either way.

Task 3.1b • Naming the force

Pulls toward Earth → **gravity / weight**

Slows a sliding box → **friction**

Pushes back on a parachute → **air resistance / drag**

Table pushing up → **normal / reaction force**

Keeps a boat afloat → **upthrust / buoyancy**

Stretched spring → **elastic / tension**

Friction and **air resistance** always oppose motion.

Task 3.3b • Quick resultants

$50 / 50 \rightarrow 0 \text{ N}$ • balanced • motion unchanged

$80 / 35 \rightarrow 45 \text{ N left}$ • speeds up left

$12 / 40 \rightarrow 28 \text{ N right}$ • speeds up right

$0 / 25 \rightarrow 25 \text{ N right}$ • speeds up right

Task 3.2 • Real situations

SITUATION	ANSWER	WHY
Book on a table	Balanced	Gravity down, table pushes up, equally
Car at steady 60 mph	Balanced	Everyone gets this wrong. Engine forward = friction and air back
Car speeding up to pass	Unbalanced	Engine force now exceeds the backward forces
Skydiver at constant speed	Balanced	Air resistance now equals weight
Still tug-of-war rope	Balanced	Both teams pull hard, but equally
Ball rolling to a stop	Unbalanced	Friction acts backward with nothing pushing forward

Task 3.3 • The tug-of-war

a. $420 - 380 = 40 \text{ N}$

b. To the **left**, toward Red

c. Blue is now $380 + 40 = 420 \text{ N}$. The forces are **balanced**, so the rope stops moving.

d. **Yes — a great deal, simply balanced.** Zero *resultant* force is not zero force. Best question on the page.

Task 3.5 • Explaining it

a. As she speeds up, **air resistance increases**. Eventually it grows until it exactly equals her weight. The forces balance, so she stops accelerating — but keeps falling at that steady speed. Gravity has not gone anywhere.

b. The bus slows but the passengers **keep moving** — nothing pushed them backward. They are not thrown forward; the bus is pulled out from under them.

The last word

Never. With no friction, no air and nothing to push against, there is no unbalanced force to change its motion — so it keeps going at the same speed forever. Accept “it never stops”. This is Newton’s first law, and students find it genuinely startling.

Reading the Periodic Table

2 MINUTES

Students read element symbols, find the atomic number and mass, work out how many protons and electrons an atom has, and discover why elements in the same column behave alike. Pure pattern-reading, no chemistry required.

You do not need to know this subject

Every answer comes from reading a box or doing one subtraction. The key shows each calculation. You do not need to know any chemistry to grade this.

Photocopy these pages only

Pages 22–24

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. A calculator is unnecessary — the arithmetic is subtraction.

WHAT TO SAY TO START

“There are about ninety naturally occurring kinds of atom, and every single thing in this room is made from them. The table you are about to read is arranged so cleverly that you can predict how an element behaves just from where it sits.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the element box panel	Everything comes from this one diagram
14–26	Task 4.1 — read the boxes	Straight reading and subtraction
26–36	Task 4.2 — symbols and names	Some symbols come from Latin
36–46	Task 4.3 — groups and patterns	The payoff task
46–50	Challenge, collect sheets	Discussion question below

Questions students will ask

“Why is sodium Na and not So?”

Its symbol comes from the Latin name, natrium. The same is true for potassium (K, kalium), iron (Fe, ferrum) and lead (Pb, plumbum).

“Is the atomic number the same as the number of protons?”

Yes, always. That is what defines the element. Change the protons and it becomes a different element entirely.

“How do I get the number of neutrons?”

Mass number minus atomic number. It is on the student page, but expect to be asked.

If they finish early

Task 4.4 asks students to predict the properties of an unknown element from its position. Push early finishers to work out why the noble gases at the far right are so unreactive.

Curriculum focus • for the regular teacher’s records

Structure of atoms; organization of the periodic table; using patterns to predict properties. Discussion question: *“The table was arranged before anybody knew what an atom was made of, and gaps in it correctly predicted elements nobody had found yet. What does that tell you about patterns in science?”*

How to read an element box

16 MIN

NAME _____

CLASS _____

DATE _____

Four pieces of information in every box

11

Na
Sodium
23

11 = atomic number • the number of **protons**

Na = the symbol • one or two letters

Sodium = the name

23 = mass number • protons + neutrons

Two rules that get you everything else:

protons = atomic number

electrons = atomic number (in a neutral atom)

neutrons = mass number – atomic number

Task 4.1 • Fill in the table

DO

ELEMENT	SYMBOL	ATOMIC NO.	MASS NO.	PROTONS	ELECTRONS	NEUTRONS
Carbon	C	6	12			
Oxygen	O	8	16			
Sodium	Na	11	23			
Iron	Fe	26	56			
Gold	Au	79	197			
Helium	He	2	4			

Which two columns are always the **same** as each other, and why? _____

Task 4.1b • Metal or non-metal?

SORT

Metals sit on the **left and middle** of the table. Non-metals sit on the **right**. Check one box for each.

ELEMENT	METAL	NON-METAL	ELEMENT	METAL	NON-METAL
Iron			Helium		
Oxygen			Gold		
Copper			Carbon		

Name two properties most metals share. _____

Symbols and where they come from

18 MIN

Task 4.2 • Match them up

MATCH

Some symbols are obvious. Others come from the element's old Latin name.

SYMBOL	ELEMENT NAME
H	
K	
Pb	
N	
Cu	

CHOOSE FROM THESE

nitrogen • lead • copper • hydrogen • potassium

Three of these symbols do not match the first letter of the English name. Why do you think that is? _____

Task 4.3 • Columns tell you how it behaves

PATTERN

Elements in the same **column** (called a group) behave in similar ways.

GROUP	NAME	HOW THEY BEHAVE
Group 1	Alkali metals	Very reactive. React strongly with water.
Group 7	Halogens	Very reactive non-metals. Often colored.
Group 8	Noble gases	Almost never react with anything at all.

a. Lithium is in Group 1. Sodium is also in Group 1. What would you predict about how sodium reacts with water? _____

b. Helium is a noble gas. Would you expect it to be safe to fill a balloon with? _____

c. Which group would you choose an element from if you needed something that would **not** react with anything? _____

Task 4.3b • Reading a formula

DECODE

A formula tells you which elements are present and how many atoms of each. H_2O means two hydrogen atoms and one oxygen atom.

FORMULA	WHICH ELEMENTS?	TOTAL ATOMS
CO_2		
$NaCl$		
CH_4		
H_2SO_4		

If there is no number after a symbol, how many atoms of it are there? _____

Predict the unknown

16 MIN

Task 4.4 • The mystery element

PREDICT

An element called **Zaporium** has been discovered. Here is its box, and it sits directly **below sodium** in Group 1.

119

Zp
Zaporium
295

- How many protons? _____
- How many electrons? _____
- How many neutrons? _____
- Predict two things about how it would behave, and say why you think so.

Task 4.5 • The big idea

CHALLENGE

When the periodic table was first arranged, there were **gaps** in it. The scientist who made it said the gaps were elements nobody had discovered yet, and predicted what they would be like. He turned out to be right.

How was that possible? What did the pattern let him do?

Task 4.6 • Rows and columns

THINK

a. Going **across** a row from left to right, what happens to the atomic number? _____

b. Elements in the same **column** behave alike. Elements in the same **row** mostly do not. What does that tell you about which direction matters more? _____

c. Hydrogen sits at the very top left, above the alkali metals, but it is a gas and not a metal at all. Why might scientists still choose to put it there?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Reading the Periodic Table

ANSWERS

Task 4.1 • The completed table

Carbon p 6 e 6 n $12 - 6 = 6$
 Oxygen p 8 e 8 n $16 - 8 = 8$
 Sodium p 11 e 11 n $23 - 11 = 12$
 Iron p 26 e 26 n $56 - 26 = 30$
 Gold p 79 e 79 n $197 - 79 = 118$
 Helium p 2 e 2 n $4 - 2 = 2$

Protons and electrons are always equal in a neutral atom, because their charges cancel out.

Task 4.2 • The symbols

H = **hydrogen**
 K = **potassium**
 Pb = **lead**
 N = **nitrogen**
 Cu = **copper**
 K, Pb and Cu come from the **Latin** names: *kalium*, *plumbum* and *cuprum*. Accept any answer that mentions another language or an older name.

Task 4.1b • Metal or non-metal

Iron **metal** • Oxygen **non-metal** • Copper **metal**
 Helium **non-metal** • Gold **metal** • Carbon **non-metal**
 Metals: shiny • conduct heat and electricity • can be bent or hammered
 • usually solid at room temperature. Any two.

Task 4.3b • Reading formulas

$\text{CO}_2 \rightarrow$ carbon, oxygen • **3 atoms**
 $\text{NaCl} \rightarrow$ sodium, chlorine • **2 atoms**
 $\text{CH}_4 \rightarrow$ carbon, hydrogen • **5 atoms**
 $\text{H}_2\text{SO}_4 \rightarrow$ hydrogen, sulfur, oxygen • **7 atoms**
 No number means **one** atom.

Task 4.3 • Groups and patterns

- Sodium should also **react strongly with water**, because elements in the same group behave alike. It does — vigorously.
 - Yes**. Noble gases almost never react, so helium will not burn or explode. That is exactly why party balloons use it.
 - Group 8, the noble gases**.
- Part b is worth drawing out: this is a real safety decision made from a pattern on a table, not from testing every gas.

Task 4.4 • Zaporium

- 119 protons
- 119 electrons
- $295 - 119 = 176$ **neutrons**
- Any two of: it would be a **metal** • it would be **very reactive** • it would **react with water**. The reason must be **because it is in Group 1, like sodium and lithium** — that reasoning is what earns the mark.

Task 4.5 • the big idea

Because the table is arranged by **pattern**, a gap tells you what belongs there before anybody has found it. The neighbors above, below and beside it reveal what it should weigh and how it should behave. Accept any answer reaching “the pattern let him predict what was missing”. Mendeleev is the scientist, if a student asks.

Where Does the Energy Go?

2 MINUTES

Students name the forms of energy, build energy transfer chains for everyday objects, and work out where the wasted energy goes. Ends with efficiency calculated as a simple percentage.

You do not need to know this subject

Every chain is in the key. The rule is that energy is never destroyed, only moved and spread out — which is stated plainly on the student page.

Photocopy these pages only

Pages 27–29

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. A calculator helps for the efficiency percentages.

WHAT TO SAY TO START

“Energy cannot be created and it cannot be destroyed. So when your phone battery goes flat, where did all that energy actually go? It did not vanish. Today you are going to track it down.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the forms of energy panel	Do the flashlight chain together
14–26	Task 5.1 — name the form	Straight recall
26–40	Task 5.2 — build the chains	The core task
40–50	Task 5.3 — wasted energy and efficiency	One percentage each
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is heat always wasted?”

Not always. In an electric heater the heat is the whole point. Energy is only ‘wasted’ when it is not the form you wanted.

“Where does the wasted energy actually go?”

It spreads out into the surroundings as heat and warms them very slightly. It is still there — just too spread out to be useful.

“Can anything be 100 percent efficient?”

Essentially no. Some energy always ends up as heat. That is why perpetual motion machines do not work.

If they finish early

Task 5.4 asks students to rank four devices by efficiency and explain the order. Push early finishers to explain why an old filament bulb is so much worse than an LED.

Curriculum focus • for the regular teacher’s records

Energy forms and transfers; conservation of energy; efficiency. Discussion question: “If energy is never destroyed, why do we talk about ‘saving’ energy?”

Energy never disappears

16 MIN

NAME _____

CLASS _____

DATE _____

The forms you need to know

Chemical

stored in food, fuel, batteries

Kinetic

anything moving

Thermal

heat

Light

from the sun, lamps, screens

Electrical

moving through wires

Sound

vibrations you can hear

Gravitational

stored by being up high

Elastic

stored in a stretched spring

Worked example • a flashlight



The energy was **never destroyed**. Some became the light you wanted. The rest became heat, which spread into the air. Add it all up and it equals what the battery started with.

Task 5.1 • Name the form

DO

WHERE THE ENERGY IS	FORM	WHERE THE ENERGY IS	FORM
A stretched rubber band		A slice of pizza	
A moving skateboard		A ringing bell	
A hot cup of tea		A lit candle flame	
A book on a high shelf		A charged phone battery	

Two of these store **more than one** form at the same time. Which, and what are they? _____

Follow the energy

20 MIN

Task 5.2 • Build the chain

TRACE

Write the energy chain for each device. Include the wasted energy at the end.

DEVICE	ENERGY CHAIN
An electric fan	
A gas stove burner	
A person riding a bike uphill	
A television	
A wind turbine	

Which one on this list produces **useful** heat rather than wasted heat? _____

Task 5.3 • Where does the waste go?

THINK

a. A laptop feels warm after an hour of use. Where did that heat come from, and where does it go next?

b. Energy is never destroyed. So why can you not collect the wasted heat and use it again? _____

Task 5.3b • It has to add up

CALCULATE

Energy is never lost. The useful energy plus the wasted energy always equals what went in.

DEVICE	ENERGY IN	USEFUL OUT	WASTED ENERGY
Electric drill	300 J	240 J of movement	
Old bulb	100 J	5 J of light	
Speaker	80 J	12 J of sound	
Kettle	2000 J	1900 J of heat in the water	

In every row, what form does the wasted energy almost always take? _____

How much actually gets used?

16 MIN

Efficiency is just a percentage

$$\text{efficiency} = \text{useful energy} \div \text{total energy} \times 100$$

A bulb uses **60 J** and gives out **3 J** of light.

$$3 \div 60 = 0.05$$

$$0.05 \times 100 = \mathbf{5\% \text{ efficient}}$$

Task 5.4 • Calculate the efficiency

CALCULATE

DEVICE	ENERGY IN	USEFUL ENERGY OUT	WORKING	EFFICIENCY
Old filament bulb	100 J	5 J of light		
LED bulb	100 J	80 J of light		
Electric motor	200 J	180 J of movement		
Gasoline car engine	500 J	125 J of movement		
Electric heater	1000 J	1000 J of heat		

a. Which device is the **most** efficient, and why is that answer a bit surprising? _____

b. An old bulb wastes 95 J. What form does that energy end up as? _____

Task 5.5 • The last word

CHALLENGE

If energy can never be destroyed, why do we talk about “saving” energy at all?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Where Does the Energy Go?

ANSWERS

Task 5.1 • Naming the form

Stretched rubber band → **elastic**

Moving skateboard → **kinetic**

Hot cup of tea → **thermal**

Book on a high shelf → **gravitational**

Slice of pizza → **chemical**

Ringing bell → **sound** (and kinetic)

Candle flame → **light and thermal**

Phone battery → **chemical**

The two with more than one form are the **candle** (light + heat) and the **bell** (sound + kinetic).

Task 5.4 • Efficiency

Filament bulb $5 \div 100 = 5\%$

LED bulb $80 \div 100 = 80\%$

Motor $180 \div 200 = 90\%$

Car engine $125 \div 500 = 25\%$

Heater $1000 \div 1000 = 100\%$

a. The **heater**, at 100%. Surprising because we usually treat heat as waste — but here heat is exactly what you wanted, so none of it is wasted. **b.** The 95 J becomes **heat**.

Task 4.6 and Task 5.3b

Task 4.6. **a.** It **increases by one** each step. **b.** The **column** matters more — that is what predicts behavior. **c.** Hydrogen has one electron like the alkali metals, so it fits the numbering pattern even though its properties are different. Accept any answer noting it fits the pattern but not the behavior.

Task 5.3b. Drill $300 - 240 = 60 \text{ J}$ • Bulb $100 - 5 = 95 \text{ J}$ • Speaker $80 - 12 = 68 \text{ J}$ • Kettle $2000 - 1900 = 100 \text{ J}$. The wasted energy is almost always **heat**.

Task 5.2 • The energy chains

DEVICE	CHAIN
Electric fan	electrical → kinetic (+ sound and heat wasted)
Gas stove burner	chemical → thermal (+ light)
Person cycling uphill	chemical → kinetic → gravitational (+ heat wasted)
Television	electrical → light + sound (+ heat wasted)
Wind turbine	kinetic → electrical (+ sound and heat wasted)

The **gas stove** produces useful heat — heat is the point of it, so it is not waste. Accept any answer that names the stove with that reasoning.

Task 5.3 • Where the waste goes

a. The heat came from **electrical energy** in the components. It spreads into the air and the desk, warming them very slightly.

b. Because it has **spread out so thinly**. The energy still exists, but it is now shared across a whole room of air at a tiny temperature rise — far too dilute to collect and do anything useful with.

Task 5.5 • the last word

We are not saving energy — we are saving **useful** energy. Once it spreads out as heat it is still there but no longer usable. "Saving energy" really means **not turning useful energy into useless energy faster than we need to**. Accept anything reaching "it becomes useless".

Break the Food Web

2 MINUTES

Students read a food web, sort organisms into producers and consumers, then work out what happens to every other population when one species is removed. The knock-on reasoning is what makes this the most discussable lesson in the pack.

You do not need to know this subject

The arrows show which way the energy travels. That single rule answers most of the lesson, and the key explains every knock-on effect.

Photocopy these pages only

Pages 32–34

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Colored pencils help for the web but are not required.

WHAT TO SAY TO START

“Every arrow on this diagram means is eaten by. Today you are going to remove one animal from it, and then work out how many other species you just affected. The number is usually larger than people expect.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the food web panel	Stress which way the arrows point
14–26	Task 6.1 — read the web	Answers come straight off the diagram
26–40	Task 6.2 — remove a species	The best discussion of the pack
40–50	Task 6.3 — build your own	Several correct answers
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Which way do the arrows go?”

Always from the food toward the eater, because that is the direction the energy travels. Grass to rabbit, not rabbit to grass. Students get this backward constantly.

“Does removing one animal really matter that much?”

Yes, and Task 6.2 shows why. Anything that ate it must find other food, which then puts pressure on species that seemed unconnected.

“What is a producer?”

Something that makes its own food from sunlight — in this web, the grass and the seeds. Everything else is a consumer.

If they finish early

Task 6.4 asks students to design a web with a named species and predict the effect of removing it. Push early finishers to explain why removing a top predator can cause plants to disappear.

Curriculum focus • for the regular teacher’s records

Ecosystems and interdependence; energy flow through food webs; cause and effect in biological systems. Discussion question: “Wolves were removed from Yellowstone and the trees disappeared. How can removing a predator kill plants?”

Follow the arrows

16 MIN

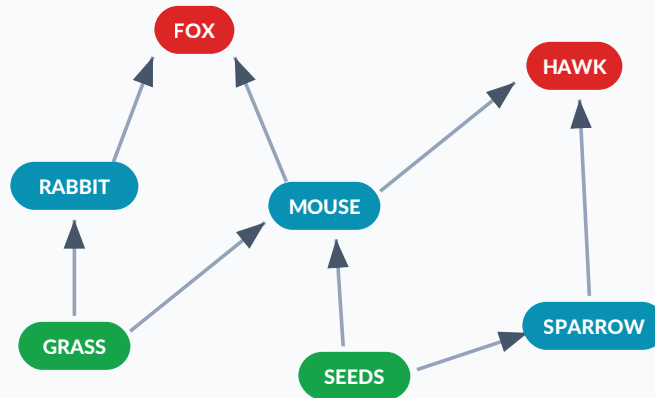
NAME _____

CLASS _____

DATE _____

An arrow means “is eaten by”

Arrows point in the direction the **energy travels**. Grass → rabbit means the rabbit eats the grass and gets its energy. Students get this backward all the time — check yours.



Producers make their own food from sunlight. **Consumers** have to eat something else.

Task 6.1 • Read the web

DO

- Name the two producers. _____
- What does the fox eat? _____
- What eats the mouse? _____
- Which animal has the most different food sources?

- Name a predator that is never eaten by anything here.

- Write one complete food chain with four links.

- Where did the energy in the grass originally come from?

Take one away

20 MIN

Task 6.2 • A disease wipes out the rabbits

PREDICT

Every rabbit in the area dies. Work out what happens next to each population, and explain **why**.

POPULATION	INCREASE	DECREASE	NO CHANGE	WHY?
Grass				
Fox				
Mouse				
Hawk				
Sparrow				

The sparrow never ate a rabbit and no rabbit ever ate a sparrow. Explain how the sparrows could still be affected.

Task 6.3 • Now add one

THINK

A new predator moves into the area. It eats **mice and sparrows** and nothing eats it.

- Which two populations fall first? _____
- What happens to the hawk, and why? _____
- What happens to the seeds, and why? _____

Task 6.3b • Sort the organisms

SORT

A **producer** makes its own food from sunlight. A **herbivore** eats only plants. A **carnivore** eats only animals. An **omnivore** eats both.

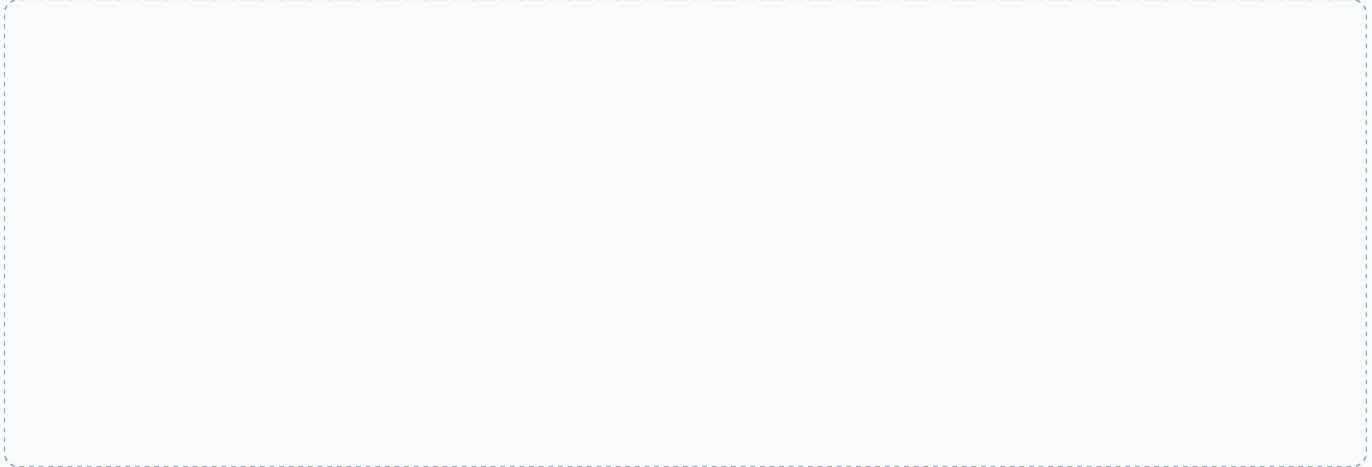
ORGANISM	WHICH ONE?	ORGANISM	WHICH ONE?
Oak tree		Seaweed	
Rabbit		Shark	
Wolf		Human	
Bear		Cow	

Every food web must start with producers. Why can a web not exist with only consumers? _____

Task 6.4 • Design an ecosystem

CREATE

Draw a food web for a pond, a desert or a rainforest. Include at least **two producers**, **three consumers** and **one top predator**. Remember which way the arrows point.



If your top predator disappeared, name two species that would be affected and say how. _____

Task 6.5 • The wolves of Yellowstone

CHALLENGE

Wolves were removed from Yellowstone National Park. Over the next few decades, whole groups of young trees disappeared along the rivers.

How can removing a **predator** cause **plants** to die? Work through the chain.

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Break the Food Web

ANSWERS

Task 6.1 • Reading the web

- a. Grass and seeds
 - b. Rabbits and mice
 - c. Foxes and hawks
 - d. The **mouse** — it eats both grass and seeds
 - e. The fox or the hawk — both are top predators
 - f. For example: grass → mouse → hawk. For four links: sun → grass → mouse → fox
 - g. The **Sun**. Grass captures sunlight, which is where every arrow on this diagram ultimately starts
- Part g is the one worth pausing on. Every food web on Earth traces back to sunlight.

Task 6.3b • Sorting the organisms

Oak tree **producer** • Rabbit **herbivore**

Seaweed **producer** • Shark **carnivore**

Wolf **carnivore** • Bear **omnivore**

Human **omnivore** • Cow **herbivore**

A web cannot exist with only consumers because **the energy has to enter from somewhere**. Only producers can capture it from sunlight. Without them there is nothing for anything else to eat.

Task 6.2 • Removing the rabbits

POPULATION	WHAT HAPPENS	WHY
Grass	Increase	One of its main eaters is gone
Fox	Decrease	Loses one of only two food sources
Mouse	Decrease	Foxes now eat mice much more heavily, since rabbits are unavailable
Hawk	Decrease	Fewer mice left for the hawk, because foxes are taking them
Sparrow	Decrease	Hawks turn to sparrows to make up for the missing mice

The sparrow question is the point of the lesson. Sparrows never met a rabbit. They are affected through **three** steps: no rabbits → foxes eat more mice → fewer mice for hawks → hawks eat more sparrows. Accept any answer that traces a chain, even if the wording is loose. Also accept “no change” for grass if a student argues mice increase to compensate — the reasoning is what you are grading.

Task 6.3 • The new predator

- a. **Mice and sparrows** — they are being eaten by something new.
- b. The **hawk decreases**. Both of its food sources are being taken by a competitor.
- c. The **seeds increase**, because both seed-eaters (mice and sparrows) have declined.

Task 6.5 • the Yellowstone chain

No wolves → **deer and elk increase** → they graze far more heavily, especially on young trees near rivers where they now feel safe standing still → **saplings are eaten before they can grow** → the tree groups disappear. Full credit for any answer with the middle step — *the animals the predator ate increased*. This really happened, and the trees returned when wolves were reintroduced.

Solid, Liquid, Gas

2 MINUTES

Students match particle diagrams to states of matter, name the six changes of state, and use the particle model to explain everyday things like a puddle drying and a balloon shrinking in the cold.

You do not need to know this subject

Everything is explained by one idea: how close the particles are and how much they move. That sentence is on the student page, and the key explains every answer from it.

Photocopy these pages only

Pages 37–39

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Nothing else.

WHAT TO SAY TO START

“Everything in this room is made of particles too small to see, and the only difference between ice, water and steam is how those particles are arranged. Today you will use that one idea to explain about eight different everyday mysteries.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the particle model panel	The three diagrams do the work
14–26	Task 7.1 — match and describe	Straight matching
26–38	Task 7.2 — name the changes	Six names, all in the key
38–50	Task 7.3 — explain the everyday	The reasoning task
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

“Is condensation the same as evaporation backward?”

Yes, exactly. Evaporation is liquid to gas, condensation is gas back to liquid.

“What is sublimation?”

Solid straight to gas without melting first. Dry ice does it, and so does frost disappearing on a cold sunny morning.

“Do the particles themselves get bigger when something is heated?”

No, and this is the most common misconception in the topic. The particles stay the same size — they just move faster and spread further apart.

If they finish early

Task 7.4 asks students to draw a heating curve. Push early finishers to explain why the temperature stops rising while ice is melting, even though heat is still going in.

Curriculum focus • for the regular teacher’s records

The particle model of matter; changes of state; using models to explain observations. Discussion question: “A puddle dries up on a day that is nowhere near 212°F. How can water become a gas without boiling?”

It is all about the particles

16 MIN

NAME _____

CLASS _____

DATE _____

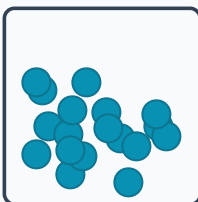
One idea explains all three states

The particles are the **same** in all three. What changes is **how close together they are** and **how much they move**.



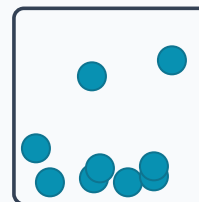
SOLID

Packed tight in a pattern. Vibrate but cannot move past each other.



LIQUID

Close together but able to slide past each other. Takes the shape of its container.



GAS

Far apart, moving fast in all directions. Fills whatever it is in.

Important: heating does not make particles bigger. It makes them **move faster**, which pushes them further apart.

Task 7.1 • Which state?

DO

DESCRIPTION	SOLID	LIQUID	GAS
Has a fixed shape you cannot change by pouring			
Fills the whole container it is put in			
Can be poured but keeps the same volume			
Particles are furthest apart			
Particles vibrate but stay in place			
Easiest to squash into a smaller space			
Why is a gas so much easier to squash than a solid? _____			

The six changes

20 MIN

Task 7.2 • Name each change

DO

Fill in the name of each change of state.

THE CHANGE	NAME	THE CHANGE	NAME
Solid → liquid		Liquid → solid	
Liquid → gas		Solid → gas	
Gas → liquid		Gas → solid	

Which three of these require energy to be **added**? _____

Task 7.3 • Explain the everyday

APPLY

Use the particle model. Every answer should mention how the particles are moving or how far apart they are.

WHAT HAPPENS	YOUR EXPLANATION
A puddle dries up on a warm day	
A cold can gets wet on the outside on a humid day	
A balloon shrinks when taken outside on a freezing day	
You can smell dinner from another room	

Task 7.3b • Draw the particles

DRAW

Draw the particles in each box. Think about how close they are and how much they would be moving.

Ice

Water

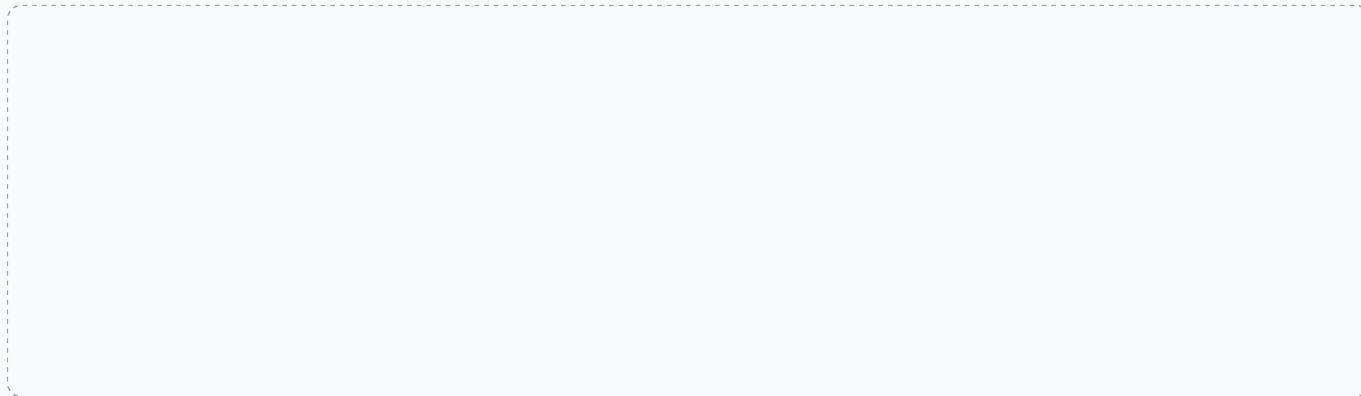
Steam

Did you draw the **same number** of particles in each box? You should have. Explain why. _____

Task 7.4 • The heating curve

DRAW

Ice at -20°C is heated steadily until it becomes steam. Sketch a graph of **temperature against time**, and label the two flat sections.



- At what temperature is the first flat section? _____
- At what temperature is the second? _____
- Heat is still going in during the flat sections, so why does the temperature stop rising?

Task 7.5 • The puddle problem

CHALLENGE

Water boils at 100°C . But a puddle dries up on a day that is only 20°C . How can water become a gas without boiling?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Solid, Liquid, Gas

ANSWERS

Task 7.1 • Which state?

Fixed shape → **solid**

Fills the container → **gas**

Pours, fixed volume → **liquid**

Particles furthest apart → **gas**

Vibrate in place → **solid**

Easiest to squash → **gas**

A gas squashes easily because there is **a lot of empty space between the particles**. In a solid they are already touching.

Task 7.2 • The six changes

Solid → liquid = **melting**

Liquid → gas = **evaporating / boiling**

Gas → liquid = **condensing**

Liquid → solid = **freezing**

Solid → gas = **sublimation**

Gas → solid = **deposition**

Energy is **added** for melting, evaporating and sublimation — the three that move toward a gas.

Task 7.3b • Drawing the particles

Ice: packed in a neat pattern, touching. Water: still touching but jumbled and able to slide. Steam: far apart and spread out. **The number of particles must be the same in all three**, because freezing or boiling water does not create or destroy any particles — it only rearranges them. A student who drew fewer particles in the steam box has the central misconception of this topic, and it is worth correcting out loud.

Task 7.3 • Explaining the everyday

WHAT HAPPENS	THE EXPLANATION
Puddle dries	Particles at the surface gain enough energy to escape as a gas. Evaporation.
Cold can gets wet	Water vapor in the air touches the cold can, loses energy, slows down and turns back to liquid. Condensation. The water came from the air, not through the can
Balloon shrinks in cold	The gas particles lose energy and move more slowly, so they hit the walls less often and less hard. The particles do not shrink
Smell travels	Gas particles from the food spread out and mix through the air. Diffusion.

The cold-can answer is worth checking carefully. Many students think the water leaked out of the can.

Task 7.4 • The heating curve

The graph rises, flattens, rises, flattens, rises again.

a. 0°C — melting

b. 100°C — boiling

c. The energy going in is being used to **break the bonds between particles** rather than to speed them up. Temperature measures how fast they move, so it stays put until the change of state is complete.

Task 7.5 • the puddle

Not all particles have the same energy. At any temperature **a few of the fastest ones at the surface** have enough energy to escape, even well below boiling point. Boiling is different — that is when particles turn to gas **throughout** the liquid, not just at the surface. Accept any answer mentioning the fastest particles escaping from the top.

Water on the Move

2 MINUTES

Students label the water cycle, trace a single water molecule through a full journey, and then read a simple weather map to make a forecast. Two topics that pair naturally and both work entirely on paper.

You do not need to know this subject

The cycle has four main stages and they are named on the student page. Weather map symbols are given in a key on the page itself. Nothing is assumed.

Photocopy these pages only

Pages 42–44

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Colored pencils are useful for the cycle diagram.

WHAT TO SAY TO START

"The water in your glass this morning has been through a dinosaur, a cloud and an ocean. There is no new water on Earth – the same water goes around and around, and today you are going to follow one drop all the way."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the water cycle panel	Four stages, named
14–26	Task 8.1 — label and define	Straight recall
26–36	Task 8.2 — follow one drop	Creative but assessable
36–46	Task 8.3 — read the weather map	Symbols given on the page
46–50	Challenge, collect sheets	Discussion question below

Questions students will ask

"What is transpiration?"

Water evaporating out of plant leaves. It is a real and large part of the cycle — a big tree releases hundreds of gallons a day.

"Why do clouds form high up?"

Air gets colder as it rises. Cold air cannot hold as much water vapor, so it condenses into droplets.

"Does the water really never run out?"

The total amount does not change, but usable fresh water in a particular place certainly can. Worth saying if it comes up.

If they finish early

Task 8.4 asks students to explain why it rains more on one side of a mountain than the other. Push early finishers to explain why coastal places have milder winters than inland ones.

Curriculum focus • for the regular teacher's records

The water cycle; the role of the Sun in driving Earth's systems; interpreting weather data. Discussion question: *"The water cycle needs energy to lift billions of gallons into the sky every day. Where does all that energy come from?"*

The same water, forever

16 MIN

NAME _____

CLASS _____

DATE _____

Four stages, driven by the Sun

EVAPORATION

The Sun heats water in oceans, lakes and rivers. It turns to vapor and rises.

CONDENSATION

High up it is colder. The vapor turns back into tiny droplets, forming clouds.

PRECIPITATION

Droplets join until they are heavy enough to fall as rain, snow or hail.

COLLECTION

Water gathers in oceans, lakes and underground, ready to start again.

There is a fifth word worth knowing: **transpiration** is water evaporating out of plant leaves. A large tree can release hundreds of gallons a day.

Task 8.1 • Match the word to the job

DO

WHAT IS HAPPENING

WHICH STAGE?

The Sun heats the surface of a lake

Water vapor cools high in the sky and forms droplets

Snow falls on a mountain

Rainwater soaks into the ground

Water escapes from the leaves of a tree

A cloud gets so heavy that rain falls

Which stage is the **only** one that needs energy added to it, and where does that energy come from? _____

Task 8.1b • Put the cycle in order

SORT

Number these 1 to 6 to show one complete journey, starting at the ocean.

ORDER

WHAT HAPPENS

Droplets join together until they are heavy enough to fall

The Sun heats the surface of the ocean

Water flows down rivers back to the sea

Water vapor rises into cooler air

Rain lands on hills and soaks into the ground

Vapor cools and condenses into a cloud

One drop's journey

20 MIN

Task 8.2 • Follow the water

WRITE

You are a single drop of water starting in the Pacific Ocean. Describe your journey through **all four stages** and back to the ocean. Use the four stage names correctly.

Task 8.3 • Read the weather map

INTERPRET

Here is a simple forecast table for one week.

DAY	HIGH °F	LOW °F	CLOUD	RAIN CHANCE	WIND
Mon	72	55	clear	5%	light
Tue	75	58	partly cloudy	10%	light
Wed	68	57	cloudy	60%	strong
Thu	61	49	cloudy	85%	strong
Fri	64	48	partly cloudy	20%	moderate
Sat	70	52	clear	5%	light

a. Which day had the largest temperature range, and what was it?

b. Which day would you choose for a picnic, and why?

c. Describe the pattern between cloud cover and temperature.

d. Rain chance on Thursday is 85%. Does that guarantee rain?

Why it rains where it does

16 MIN

Task 8.4 • The mountain problem

THINK

Wind blows in off the ocean and hits a mountain range. The side facing the ocean is green and rainy. The far side is dry, sometimes desert.

OCEAN → / \ → DRY SIDE
wet side / \

a. What happens to air when it is forced upward over a mountain?

b. Why does that cause rain on the ocean side?

c. By the time the air reaches the far side, what has it lost? _____

Task 8.5 • The big question

CHALLENGE

Every day the water cycle lifts billions of gallons of water into the sky. That takes an enormous amount of energy.

Where does all that energy come from, and what would happen to the cycle without it?

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

Water on the Move

ANSWERS

Task 8.1 • Matching

Sun heats a lake → **evaporation**

Vapor cools, forms droplets → **condensation**

Snow falls → **precipitation**

Rain soaks into the ground → **collection**

Water leaves a tree → **transpiration**

Heavy cloud drops rain → **precipitation**

Evaporation is the stage needing energy, and it comes from **the Sun**.

Task 8.2 • Grading the journey

Full credit needs all four stage names used **correctly and in order**: evaporation from the ocean, condensation into a cloud, precipitation as rain or snow, collection in a river or lake, then back to the ocean. Style and creativity are not being graded — the vocabulary and the sequence are.

Task 8.1b • The cycle in order

- 1 The Sun heats the surface of the ocean
- 2 Water vapor rises into cooler air
- 3 Vapor cools and condenses into a cloud
- 4 Droplets join until heavy enough to fall
- 5 Rain lands on hills and soaks into the ground
- 6 Water flows down rivers back to the sea

Task 8.3 • Reading the forecast

- a. **Monday**, range $72 - 55 = 17^{\circ}\text{F}$. Tuesday is also 17, so accept either — a student who spots the tie has read carefully.
- b. **Monday or Saturday** — clear, warm, 5% rain, light wind. Any answer with a reason drawn from the table.
- c. **Cloudier days are cooler**. Clouds block sunlight from reaching the ground.
- d. **No**. 85% means it is very likely, not certain. Roughly speaking, on 100 days like Thursday it would rain on about 85 of them.

Task 8.4 • The mountain

- a. It **rises and cools**.
- b. Cold air cannot hold as much water vapor, so it **condenses** into cloud and falls as rain on the ocean side.
- c. It has lost **most of its moisture**, so the far side gets little rain. This is called a rain shadow.

Task 8.5 • the big question

All of it comes from **the Sun**. Without solar energy there would be no evaporation, so no clouds, no rain, and no cycle at all — the water would simply sit in the oceans. Accept any answer naming the Sun. It is worth stating explicitly that the Sun drives essentially every weather system on Earth.

A Restless Planet

2 MINUTES

Students label Earth's four layers, learn the three kinds of plate boundary, and then use a map description to explain why earthquakes and volcanoes appear in lines rather than scattered randomly.

You do not need to know this subject

Four layers and three boundary types, all named on the student page with what happens at each. The key explains every answer.

Photocopy these pages only

Pages 47–49

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. Colored pencils help with the layers diagram.

WHAT TO SAY TO START

"Earthquakes do not happen randomly. If you plot every one from the last century on a map, they form neat lines across the planet. Today you will find out what those lines actually are."

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the layers panel	Four layers, outside to inside
14–24	Task 9.1 — label the layers	Straight recall
24–38	Task 9.2 — the three boundaries	The core of the lesson
38–50	Task 9.3 — why the lines	The reasoning task
50–55	Challenge, collect sheets	Discussion question below

Questions students will ask

"Is the mantle liquid?"

Not quite. It is solid rock that flows extremely slowly, like very thick tar. Only the outer core is genuinely liquid.

"How fast do the plates move?"

About as fast as your fingernails grow — a few centimeters a year. Over millions of years that adds up to entire oceans.

"Why is the inner core solid if it is hotter?"

Because the pressure is enormous. Pressure can keep something solid even above its normal melting point.

If they finish early

Task 9.4 asks students to predict what will happen to a named place over millions of years. Push early finishers to explain why the Himalayas are still getting taller.

Curriculum focus • for the regular teacher's records

Earth's internal structure; plate tectonics; using patterns to explain the distribution of geological events. Discussion question: *"If the plates have always moved, what did the map of the world look like 200 million years ago?"*

Four layers down

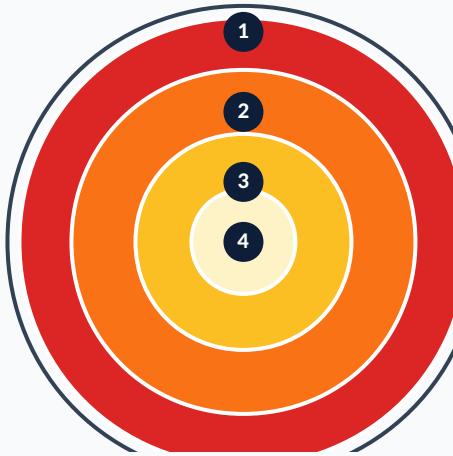
16 MIN

NAME _____

CLASS _____

DATE _____

From the surface to the center



LAYER

WHAT IT IS LIKE

Crust

Thin, solid rock. Everything we live on. Only about 5–40 miles thick.

Mantle

Thick, hot rock that flows extremely slowly. The biggest layer by far.

Outer core

Liquid metal, mostly iron. Its movement makes Earth's magnetic field.

Inner core

Solid metal, hotter than the surface of the Sun, kept solid by enormous pressure.

Task 9.1 • Label and answer

DO

#

LAYER NAME

1

2

3

4

a. Which layer is the thinnest? _____

b. Which is the only truly liquid layer? _____

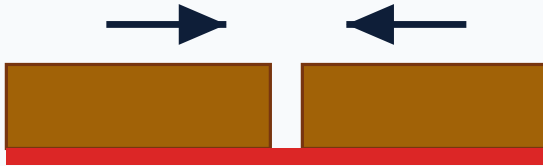
c. The inner core is hotter than the outer core but is solid. Why? _____

Where the plates meet

20 MIN

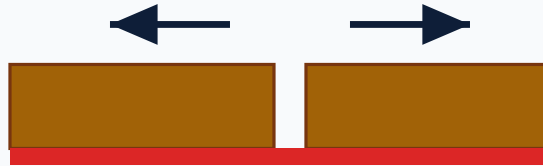
The crust is cracked into moving pieces

Earth's crust is broken into huge slabs called **plates** that float on the mantle and move a few centimeters a year — about as fast as your fingernails grow. Almost all earthquakes and volcanoes happen where two plates meet.



CONVERGENT

Pushing together. Makes mountains, volcanoes and huge earthquakes.



DIVERGENT

Pulling apart. New crust forms in the gap. Mostly under oceans.



TRANSFORM

Sliding past each other.

Task 9.2 • Which boundary?

DECIDE

WHAT IS HAPPENING THERE

BOUNDARY TYPE

WHY?

Two plates crush together and push up a mountain range

New seafloor is being created in the middle of an ocean

Two plates grind past each other sideways

A chain of volcanoes runs along a coastline

A long crack where a continent is slowly splitting apart

Which boundary type produces earthquakes but almost no volcanoes? _____

Task 9.2b • How far, how long?

CALCULATE

Two plates are moving apart at **3 cm per year**.

a. How far apart in 100 years? _____

b. How far in 1,000 years? _____

c. How far in 1 million years, in kilometers? _____

d. The Atlantic Ocean is about 5,000 km wide. Roughly how long would it take to open at this rate? Show your thinking.

Task 9.3 • The pattern on the map

EXPLAIN

If you mark every earthquake of the last hundred years on a world map, they do not scatter evenly. They form **long thin lines**, especially in a huge ring around the Pacific Ocean.

- What do those lines actually show? _____
- Japan and Chile are on opposite sides of the world and both have frequent earthquakes and volcanoes. What do they have in common?

- The middle of Australia has almost no earthquakes at all. What does that tell you about where it sits?

Task 9.4 • Millions of years from now

CHALLENGE

- India is still pushing north into Asia. What is happening to the Himalayas because of it? _____
- The plates have been moving for billions of years. What would a world map have looked like 200 million years ago?

Task 9.5 • True or false?

DECIDE

STATEMENT	TRUE	FALSE	IF FALSE, CORRECT IT
The crust is the thickest layer.			
Plates move about as fast as fingernails grow.			
Earthquakes happen randomly all over the world.			
The inner core is liquid.			
New crust is made at divergent boundaries.			

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

Answer key

A Restless Planet

ANSWERS

Task 9.1 • The four layers

1 Crust • 2 Mantle

3 Outer core • 4 Inner core

- a. The **crust** — by an enormous margin.
- b. The **outer core**. The mantle is solid rock that flows very slowly, which is not the same thing.
- c. The **pressure** is so great that it forces the metal to stay solid even at that temperature.

Task 9.2 • Which boundary?

Mountains pushed up → **convergent**New seafloor forming → **divergent**Grinding past sideways → **transform**Volcano chain on a coast → **convergent**Continent splitting → **divergent**

Transform boundaries give earthquakes but almost no volcanoes, because nothing is melting — the plates are only scraping.

Task 9.2b • How far, how long

- a. $3 \times 100 = 300$ cm (3 m)
- b. $3 \times 1,000 = 3,000$ cm (30 m)
- c. $3 \times 1,000,000 = 3,000,000$ cm = 30 km
- d. $5,000 \div 30$ km per million years
 ≈ 170 million years

Close to the real figure, which is why this calculation is worth doing.

Task 9.5 • True or false

Crust is thickest → **False**. It is the thinnestPlates move like fingernails → **True**Earthquakes are random → **False**. They follow plate boundariesInner core is liquid → **False**. Solid; the outer core is liquidNew crust at divergent → **True**

Task 9.3 • The pattern

- a. They show **the edges of the tectonic plates**. Earthquakes happen where plates meet, so plotting earthquakes effectively draws the plate boundaries.
- b. Both sit **on the edge of the Pacific plate**, on the ring of boundaries known as the Ring of Fire. Being on opposite sides of the world is irrelevant — what matters is being on a boundary.
- c. Central Australia is in the **middle of a plate**, far from any boundary, so there is very little to shake it.
Part a is the key insight and worth saying aloud: **the map of earthquakes is a map of the plate edges**.

Task 9.4 • the future and the past

- a. The Himalayas are still being pushed **higher** — roughly half an inch a year, though erosion removes some of it. b. Very different. The continents were joined into one or two enormous landmasses; the supercontinent is called **Pangaea**. A strong clue students often notice by themselves is that South America and Africa look like they would fit together — because they once did. Accept any answer saying the continents were in different places or joined together.

Do the Numbers Agree?

2 MINUTES

Students read a results table and a line graph, decide what the data does and does not show, then evaluate four advertising claims against the evidence behind them. The lesson that makes every other science lesson useful.

You do not need to know this subject

There is no content to know. Every question is about whether the numbers actually support what somebody said, and the key explains each one.

Photocopy these pages only

Pages 52–54

One per student. **Do not photocopy this page or the answer key.**

What students need

A pencil. A calculator is not needed.

WHAT TO SAY TO START

“Here are four claims from real advertisements. Every one of them is technically true, and every one of them is designed to make you believe something the evidence does not actually show. Today you learn to tell the difference.”

HOW THE LESSON RUNS

TIME	WHAT HAPPENS	WHAT YOU DO
0–5	Hand out sheets, settle	Use the door script on page 5
5–14	Read the checklist panel	Four questions to ask of any claim
14–28	Task 10.1 – read the graph	Answers come off the graph
28–42	Task 10.2 – what does it NOT show	The valuable one
42–54	Task 10.3 – judge the claims	Best discussion in the pack
54–60	Challenge, collect sheets	Discussion question below

Questions students will ask

“Can’t we just believe the numbers?”

The numbers are usually true. It is the **conclusion drawn from them** that is often wrong. That distinction is the whole lesson.

“How many results are enough?”

There is no fixed number, but two is never enough and more is almost always better. Ask what would happen if one result were a fluke.

“Isn’t ‘four out of five’ good evidence?”

Only if you know how many were asked, who they were, and who paid for the study. Task 10.3 makes students ask exactly that.

If they finish early

Task 10.4 asks students to write a misleading-but-true claim of their own. Push early finishers to explain what extra information a reader would need to see through it.

Curriculum focus • for the regular teacher’s records

Analyzing and interpreting data; distinguishing evidence from conclusion; evaluating scientific claims. Discussion question: “Why does it matter who paid for a study, if the numbers in it are true?”

Four questions to ask

16 MIN

NAME _____

CLASS _____

DATE _____

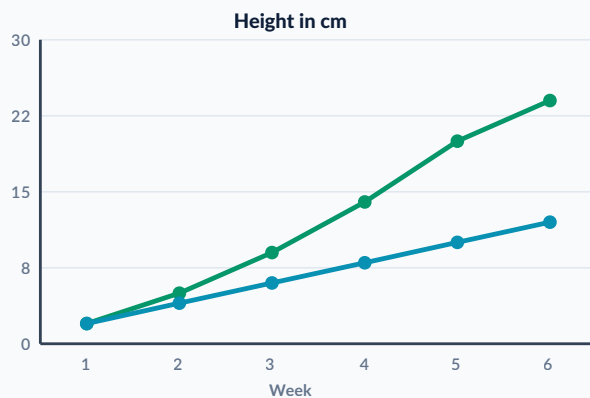
Before you believe any result

- 1 **How many were tested?** Two plants prove nothing. Two hundred is harder to argue with.
- 2 **Was it a fair test?** Was only one thing changed?
- 3 **Does the conclusion match the data?** The numbers may be true while the conclusion is not.
- 4 **Who paid for it?** Not proof of anything by itself — but always worth knowing.

Task 10.1 • Read the graph

DO

Two tomato plants were grown for six weeks. Plant A was given a new fertilizer. Plant B was not.



- a. How tall was Plant A at week 4? _____
- b. How tall was Plant B at week 6? _____
- c. What was the difference between them at week 6?

- d. Between which two weeks did Plant A grow fastest?

- e. Describe the shape of Plant B's line in one sentence.

Task 10.1b • Read the results table

DO

Four students each dropped the same ball from the same height onto four surfaces and measured the bounce.

SURFACE	TRY 1	TRY 2	TRY 3	AVERAGE
Concrete	62 cm	64 cm	60 cm	
Wood	55 cm	53 cm	54 cm	
Carpet	31 cm	29 cm	30 cm	
Grass	18 cm	41 cm	19 cm	

- a. Which surface gave the highest bounce? _____
- b. One result looks wrong. Which one, and what should the students do about it? _____

What the data cannot tell you

20 MIN

Task 10.2 • Does the conclusion hold up?

JUDGE

The fertilizer company writes: "Our fertilizer doubles the growth of tomato plants!"

QUESTION

YOUR ANSWER

How many plants were tested altogether?

Is that enough to be confident? Why?

Name two things we are not told about how the plants were grown

Does the graph actually show growth being doubled?

Write a fairer version of the company's claim

The **numbers on the graph are completely true**. Explain how a true graph can still lead to a misleading claim.

Task 10.2b • Evidence or opinion?

SORT

Evidence can be checked by measuring. Opinion cannot.

THE STATEMENT

EVIDENCE

OPINION

HOW DO YOU KNOW?

Plant A reached 24 cm in six weeks.

This is the best fertilizer available.

Plant A grew twice as tall as Plant B.

Gardeners will love these results.

The plants were grown for six weeks.

Advertisements usually mix the two together. Why does that work so well on people? _____

Judge the claims

18 MIN

Task 10.3 • What would you need to know?

INVESTIGATE

Every claim below could be technically true. For each, write the **one question** you would most want answered before believing it.

THE CLAIM

THE QUESTION I WOULD ASK

"4 out of 5 dentists recommend this toothpaste."

"Clinically proven to reduce wrinkles."

"This cereal is 30% less sugar."

"Studies show our vitamin boosts energy."

Task 10.4 • Write a misleading truth

CHALLENGE

Invent a product and write a claim about it that is **completely true** but leaves people with the wrong impression.

What information would a reader need in order to see through it? _____

Before you hand this in

CHECKPOINT

Tick what you managed today

- ☐ I finished the main tasks
- ☐ I attempted the stretch task
- ☐ I checked my work with a partner

One thing I now understand that I did not this morning

Teacher initials _____

The last word. If the numbers in a study are true, why does it matter who paid for it?

Answer key

Do the Numbers Agree?

ANSWERS

Task 10.1 • Reading the graph

- a. Plant A at week 4 = **14 cm**
- b. Plant B at week 6 = **12 cm**
- c. $24 - 12 = 12$ **cm**
- d. Between weeks **4 and 5** (14 to 20, a jump of 6)
- e. A **straight line** — Plant B grew steadily, about 2 cm every week.

Task 10.2 • the key answers

Two plants — nowhere near enough, because one plant could simply be a naturally strong one. Not told: pot size, soil, water, light, temperature, plant variety, starting height. And **yes, growth was doubled** (24 vs 12) — but only for these two plants, in one test, run by the company selling it.

Task 10.1b • The bounce table

Concrete avg **62 cm** • Wood avg **54 cm**

Carpet avg **30 cm** • Grass → see below

a. Concrete. b. The **41 cm grass result** is the odd one out — the other two are 18 and 19. It should be **repeated**, and if it stays odd, investigated rather than silently deleted. Ignoring it gives an average of about 18.5 cm.

Task 10.2b • Evidence or opinion

Reached 24 cm → **evidence**

Best available → **opinion**

Twice as tall as B → **evidence**

Gardeners will love it → **opinion**

Grown for six weeks → **evidence**

Mixing them works because the true measurements **lend their credibility** to the opinions sitting beside them.

Task 10.2 • A fairer claim, and the big idea

A fairer version: “In one small trial with two plants, the plant given our fertilizer grew twice as tall over six weeks.” Accept any wording that adds the sample size or the limits of the test.

How a true graph misleads: the graph shows **what happened to two particular plants**. The claim implies **what will happen to your plants**. That leap — from a small result to a general promise — is where the misleading happens, and no number on the graph is false. This is the single most important idea in the lesson.

Task 10.3 • The questions to ask

THE CLAIM	THE BEST QUESTION
4 out of 5 dentists	How many dentists were asked? If it was five, the claim means four people. Also: who asked them, and were they paid?
Clinically proven	Proven by whom, on how many people, and compared with what? “Clinically proven” is not a regulated phrase.
30% less sugar	Less than what? Less than their own previous version? Less than cake? It may still be very high in sugar.
Studies show	Which studies, and who funded them? “Studies show” with no named study is not evidence at all.

Any sensible question earns the mark. The pattern to draw out: each claim **leaves out the number or the comparison** that would let you judge it.

The last word

Funding does not make results false, and students should not conclude that it does. But somebody who wants a particular answer chooses what to test, what to compare against, and **which results to publish**. A company can run twenty trials and report only the one that worked. Knowing who paid tells you what questions to ask — it is not proof of anything on its own.

Handover note

90 SECONDS

Copy this page or fill it in directly. Ninety seconds of your time saves the regular teacher a great deal of guesswork.

SUBSTITUTE
_____CLASS
_____DATE
_____PERIOD

Lesson used

CHECK ONE

- | | |
|--|---|
| ☐ 01 • Think Like a Scientist | ☐ 06 • Break the Food Web |
| ☐ 02 • Inside a Cell | ☐ 07 • Solid, Liquid, Gas |
| ☐ 03 • Push, Pull, Go | ☐ 08 • Water on the Move |
| ☐ 04 • Reading the Periodic Table | ☐ 09 • A Restless Planet |
| ☐ 05 • Where Does the Energy Go? | ☐ 10 • Do the Numbers Agree? |

How far did the class get?

- ☐ Everyone finished the main tasks
- ☐ Most finished; some are partway
- ☐ We got through about half
- ☐ We did not get far — see notes

Sheets

- ☐ Collected and left on the desk
- ☐ Students kept them
- ☐ Graded against the answer key

Anything the regular teacher should know

NOTES

Students to mention • for any reason, good or otherwise

NAMES

Thank you

Walking into an unfamiliar subject is not easy. Leaving a clear note behind is what makes the next teacher's day work, and it is genuinely appreciated.

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Standards alignment

Written to the Grade 6–8 middle school science standards used across the United States. Across the ten lessons the pack covers:

- Planning investigations and controlling variables
- Structure and function of cells
- Forces, motion and Newton's laws
- Elements, atoms and the periodic table
- Energy transfer and conservation
- Ecosystems, food webs and interdependence
- The particle model and changes of state
- The water cycle and weather
- Earth's structure and plate tectonics
- Analyzing and interpreting data

Maps readily to NGSS middle school performance expectations across all three domains. Individual teacher pages name the focus for that lesson.

Also from Digital Vector

Emergency Sub Plans • Middle School Math

The same format for a math class. Ten no-prep lessons, worked answer keys.

Emergency Sub Plans • Digital Technologies

The same format for a computer science or technology class.

AI Explorers & AI Literacy

How AI works, where it fails, and how to use it honestly — for Grades 3–5 and 5–9.

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Found an error, or want a topic that is not here?

Feedback is genuinely read. If a lesson did not land with your class, or you need a sub plan on a topic this pack does not cover, get in touch through digitalvector.com.au and it will be considered for the next update.

EMERGENCY SUB PLANS PACK

Middle School Science • Grades 6–8

Ten lessons • No prep • No lab • No equipment